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Assignment 2

Expected values + option.

A stock has a currently selling price of \$20. The Distribution of the price of the stock one year from now is believed to be as follow.

Price (x)	Probability (fx)
20	0.05
21	0.06
22	0.07
23	0.08
24	0.09
25	0.12
26	0.15
27	0.13
28	0.10
29	0.10
30	0.05
	1

stock price can be 20-30

option value = max (stock price − 25, 0)

If the stock is below 25 → option is worth 0
If above 25 → you make (stock − 25)

1.) mean $\mu = \sum (x \times P)$

multiple each Price by it's probability + add.

$20(.05) + 21(.06) + 22(.07) + 23(.08) + 24(.09) + 25(.12) + 26(.15) + 27(.13) + 28(.10) + 29(.10) + 30(0.5) = 1$

(mean stock price) $\mu = 25.41$

how spread out prices are

→ SD $\sigma = \sqrt{\sum (x - \text{mean})^2 \times P}$

$[(20-25.4)^2 \times 0.05) + (21-25.41)^2(0.06) + (22-25.41)^2(0.07) + (23-25.41)^2(0.08) + (24-25.41)^2 \times (0.09) + (25-25.41)^2(0.12) + (26-25.41)^2(0.15) + (27-25.41)^2(0.13) + (28-25.41)^2 \times (0.10) + (29-25.41)^2(0.10) + (30-25.41)^2(0.05) =$

$[1.46 + 1.166 + 0.813 + 0.464 + 0.178 + 0.0492 + 0.0885 +$

SD = 2.74